# MTG Satellites, Instruments and Data, a limited overview

NOA/ReACT

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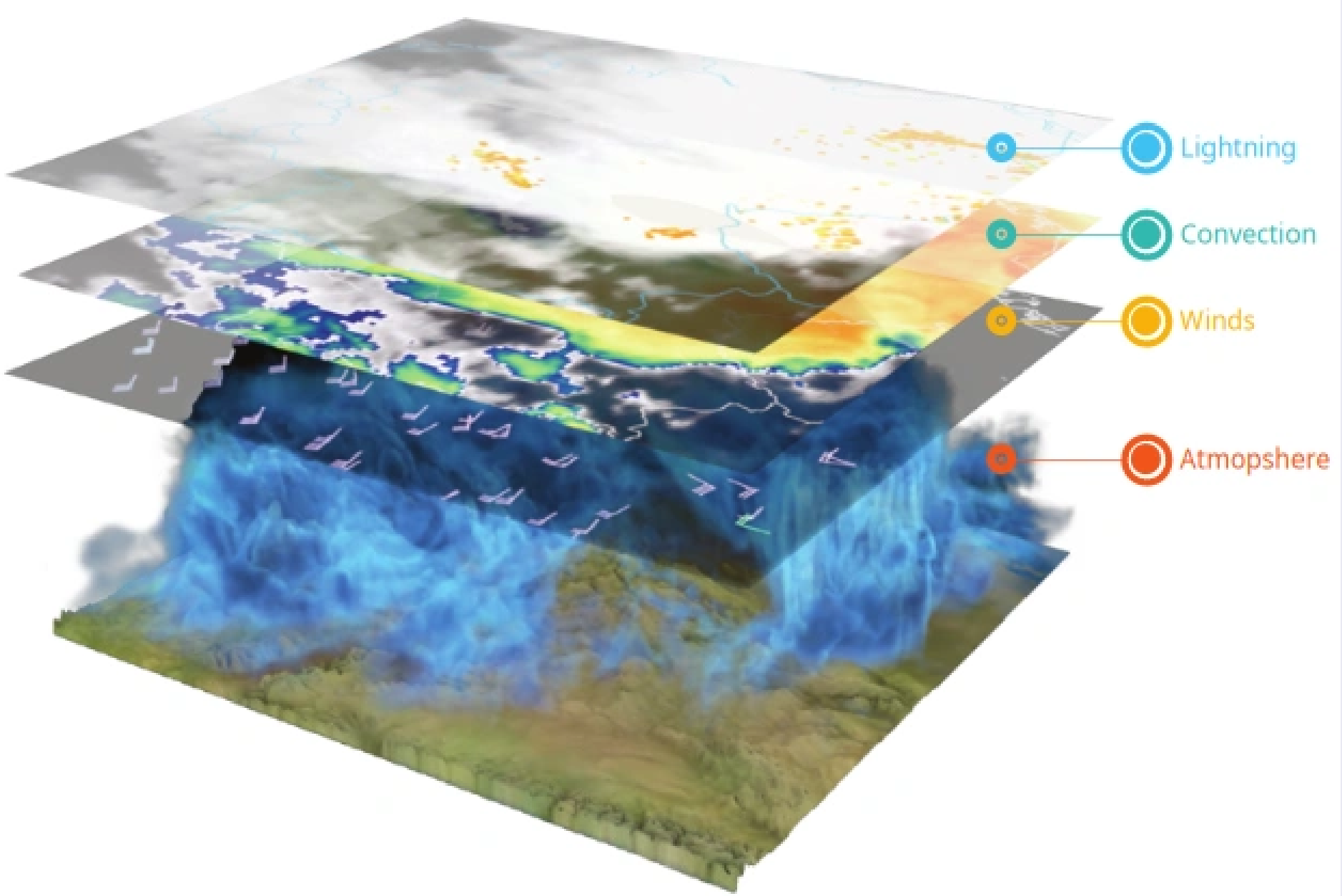
2026-02-25

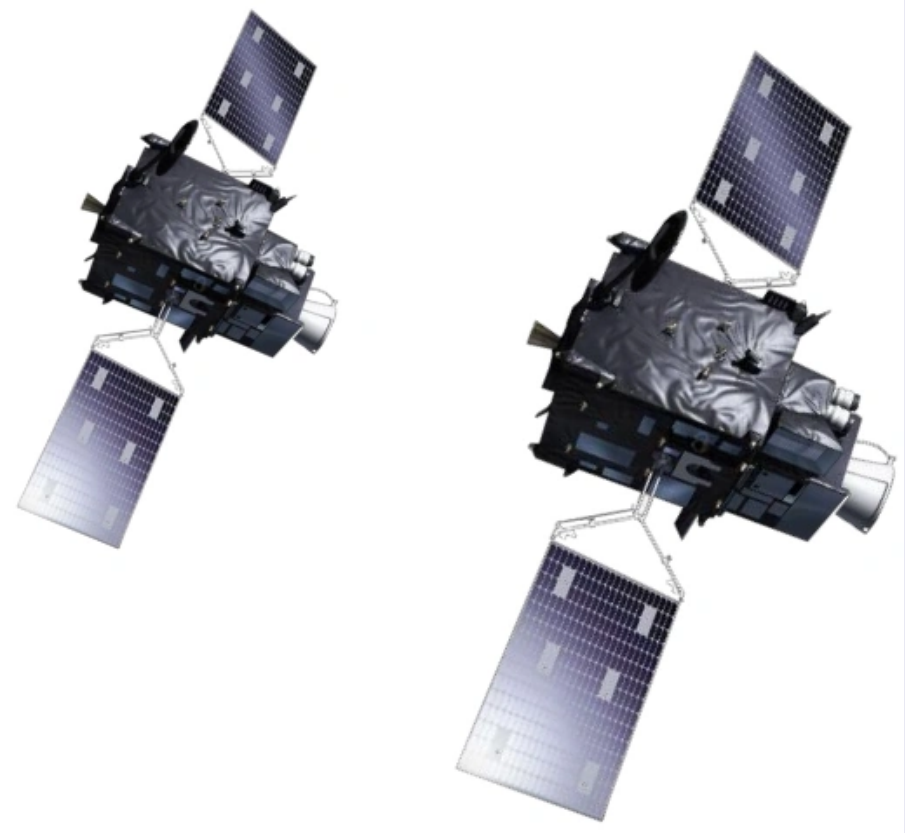
## Definitions

- **MTG**: Meteosat Third Generation system of geostationary satellites operated by EUMETSAT.
- **EUMETSAT**: Europe's meteorological satellite agency, an intergovernmental organization.
  - **goal**: Produce satellite data on weather, climate, and the environment. Provide near-real time data for members, partners, and other data users.
  - **fields**: Nowcasting, Numerical Weather Prediction models, Climate records, Other products and services (firefighting, air quality forecasts, air traffic control, search and rescue missions, disaster risk reduction, agricultural productivity, marine and coastal management.....)

# Data scope



# MTG-I Imaging mission



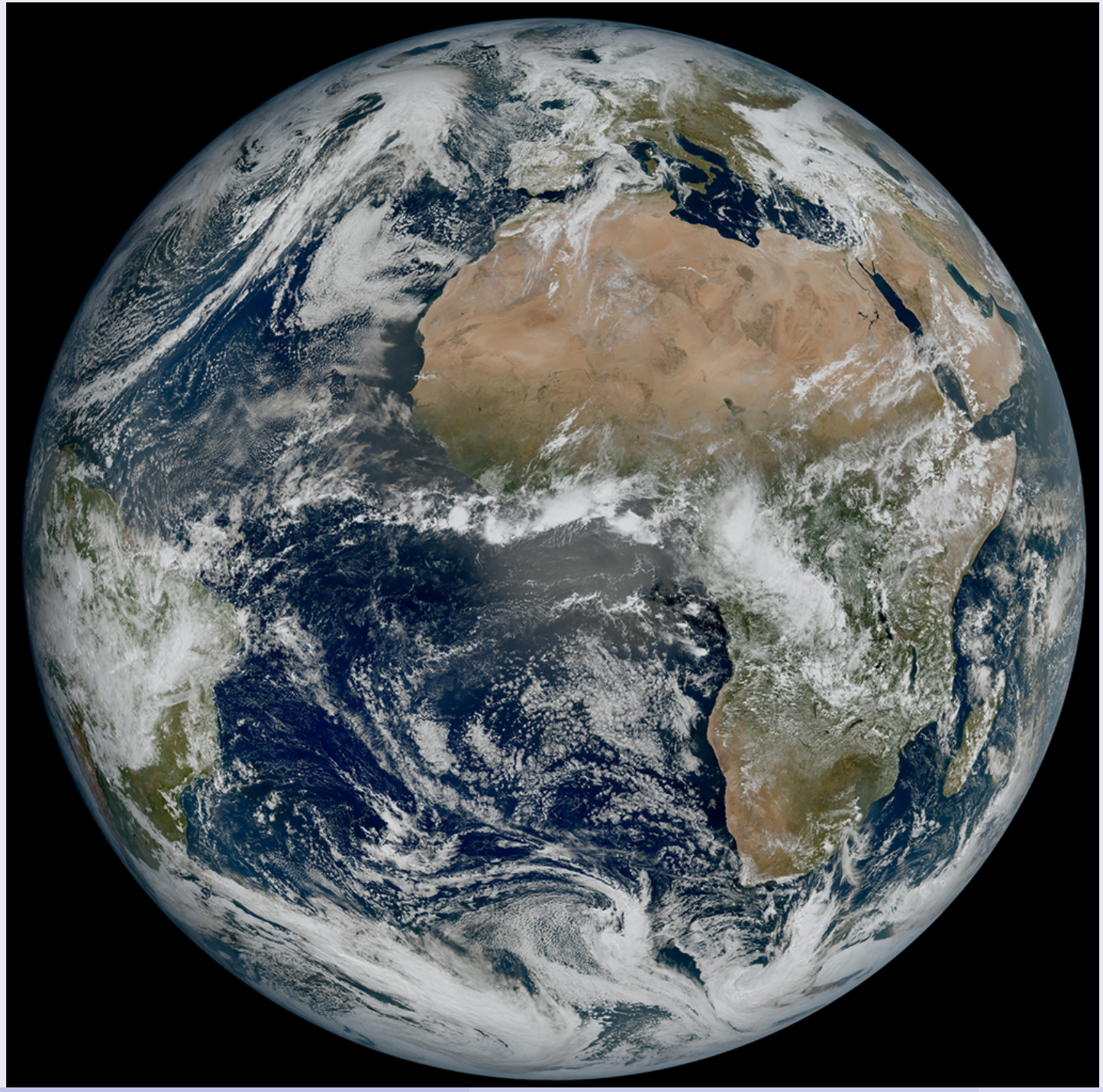
Σχήμα 2: MTG-I

- *FCI* - Full disc scanning service
  - /10 minutes, 16 spectral bands
- *FCI* - Rapid scanning service
  - Europe /2.5 minutes, 16 spectral bands
- *LI* - Lightning Imager
  - Full disc, continuous

www.eumetsat.int/meteosat-third-generation
user.eumetsat.int/resources/user-guides/mtg-in-operations

# MTG-S Sounding mission

- Infrared Sounder (IRS)
  - 1/4 of Full disk /15 minutes
- Ultraviolet Visible Near-infrared (UVN) Sounder
  - Copernicus Sentinel-4
  - Smaller disc than FCI, /60min



# Flexible Combined Imager FCI

Full Earth disk every 10 minutes

| | Channel | λc | Δλ | SSD |
|---|---|---|---|---|
| 1 | VIS 0.4 | 0.444 µm | 0.060 µm | 1.0 km |
| 2 | VIS 0.5 | 0.510 µm | 0.040 µm | 1.0 km |
| 3 | VIS 0.6 | 0.640 µm | 0.050 µm | 1.0 km 0.5 km (HR) |
| 4 | VIS 0.8 | 0.865 µm | 0.050 µm | 1.0 km |
| 5 | VIS 0.9 | 0.914 µm | 0.020 µm | 1.0 km |
| 6 | NIR 1.3 | 1.380 µm | 0.030 µm | 1.0 km |
| 7 | NIR 1.6 | 1.610 µm | 0.050 µm | 1.0 km |
| 8 | NIR 2.2 | 2.250 µm | 0.050 µm | 1.0 km 0.5 km (HR) |
| 9 | IR 3.8 | 3.800 µm | 0.400 µm | 2.0 km 1.0 km (HR) |
| 10 | WV 6.3 | 6.300 µm | 1.000 µm | 2.0 km |
| 11 | WV 7.3 | 7.350 µm | 0.500 µm | 2.0 km |
| 12 | IR 8.7 | 8.700 µm | 0.400 µm | 2.0 km |
| 13 | IR 9.7 | 9.660 µm | 0.300 µm | 2.0 km |
| 14 | IR 10.5 | 10.500 µm | 0.700 µm | 2.0 km 1.0 km (HR) |
| 15 | IR 12.3 | 12.300 µm | 0.500 µm | 2.0 km |
| 16 | IR 13.3 | 13.300 µm | 0.600 µm | 2.0 km |

*λc: Central wavelength [µm], Δλ: Spectral width [µm], SSD: Spatial Sampling Distance [km] Channels: VIS (Visible), NIR (Near IR), IR (Infrared), WV (Water Vapour)*

https://user.eumetsat.int/resources/user-guides/mtg-fci-level-1c-data-guide

# EUMETSAT MTG available products (2026-02-01)

- Accumulated precipitation at ground by blended MW+IR and FCI
- Active Fire Monitoring (CAP, netCDF)
- All Sky Radiance (BUFR, netCDF)
- Atmospheric Motion Vectors (BUFR, netCDF)
- Clear Sky Reflectance Map
- Cloud Mask (GRIB2, netCDF)
- Cloud Top Temperature and Height
- Cloud Type
- FCI Level 1c High Resolution Image Data
- FCI Level 1c Normal Resolution Image Data
- FCI Level 3 Sea Surface Temperature
- FCI Snow detection (snow mask) by VIS/NIR radiometry
- Global Instability Indices
- LI Accumulated Flash Area
- LI Accumulated Flash Radiance
- LI Accumulated Flashes
- LI Lightning Events Filtered
- LI Lightning Flashes
- LI Lightning Groups
- Land Surface Temperature
- Optimal Cloud Analysis
- Outgoing LW radiation at TOA
- Precipitation rate at ground by blended FCI IR / LEO MW precipitation

https://data.eumetsat.int/extended?query=mtg

# FCI composites (MTG-I)

Predefined composites of wavelengths...

```
from satpy import Scene
scn = Scene(filenames = files, reader = 'fci_l1c_nc')
print(scn.available_composite_names())
```

24h_microphysics, airmass, ash, cimss_cloud_type, cimss_cloud_type_raw, cloud_phase, cloud_phase_distinction, cloud_phase_distinction_raw, cloud_phase_raw, cloud_phase_with_night_ir105, cloud_type, cloud_type_with_night_ir105, cloudtop, colorized_ir_clouds, convection, day_essl_colorized_low_level_moisture, day_essl_low_level_moisture, day_microphysics, day_severe_storms, day_severe_storms_tropical, **dust**, essl_colorized_low_level_moisture, essl_low_level_moisture, fci_fire_channels_sum, fire_temperature, fire_temperature_38refl, fire_temperature_rad, flames_masked, fog, geo_color, geo_color_background_with_low_clouds, geo_color_high_clouds, geo_color_low_clouds, geo_color_night, green_snow, hrv_clouds, ir108_3d, ir_cloud_day, ir_sandwich, ir_sandwich_with_night_colorized_ir_clouds, **natural_color**, natural_color_raw, ndvi_hybrid_green, ndvi_hybrid_green_fully_sunzencorrected, ndvi_hybrid_green_raw, night_fog,

# LI Lightning Imager (MTG-I)

LI Lightning Imager Groups

| Variable | Description |
|---|---|
| `group_time` | UTC time of the group |
| `latitude` | Avg latitude of the events in the group weighted by the event radiance |
| `longitude` | Avg longitude of the events in the group weighted by the event radiance |
| `radiance` | Total radiance of the group (Optical Characteristics of OTD Flashes and the Implications...) |
| `number_of_events` | Number of events in the group |
| `flash_id` | ID of the flash that allows one to relate the groups in the LI-2-LGR-x-FD-x to the LI-2-LFL-x-FD-x |
| `group_quality_flag` | Quality of the group computed at the group-rejection step (Rejection of groups prior to the comp...) |

Key variables in LI-2-LGR-x-FD-x.

https://user.eumetsat.int/resources/user-guides/mtg-li-level-2-data-guide

# LI Lightning Imager (MTG-I)

LI Lightning Imager Flashes

| Variable | Description |
|---|---|
| `flash_time` | UTC time of the first group in the flash |
| `flash_duration` | Duration of the flash, ie, time difference between the first a the last group of the flash |
| `latitude` | Average latitude of the events in the flash weighted by the event radiance |
| `longitude` | Average longitude of the events in the flash weighted by the event radiance |
| `radiance` | Total radiance of the flash (Optical Characteristics of OTD Flashes and the Implications...) |
| `number_of_events` | Number of events in the flash |
| `number_of_groups` | Number of groups in the flash |
| `flash_footprint` | Size of the flash footprint in number of pixels |
| `flash_id` | ID of the flash |
| `flash_filter_confidence` | Quality of the flash computed at the group-rejection step (Rejection of flashes) |

## InfraRed Sounder IRS (MTG-S)

**Hyperspectral infrared sounder**

- Goal:
  - Derive the evolution of vertically resolved water vapour structures
  - Extract 3D wind fields with a vertical resolution ~ 2 km
  - Track the 3D structures
    - atmospheric water vapour and temperature
    - carbon monoxide and pollution
    - atmospheric dynamic variables
- Spatial: **4km** at nadir (1.025° x 1.025°)
- Temporal: 15 – 30 minutes for local zones, **60 minutes** for full disk

## InfraRed Sounder IRS (MTG-S)

| Band | Wavenumber Range | Channels | Main Targets |
|---|---|---|---|
| Long Wave IR (LWIR) | 679.5 – 1210.0 $cm^{-1}$ | 887 | H2O, CO2, O3, Surface, Clouds, Aerosols |
| Mid Wave IR (MWIR) | 1600.0 – 2250.4 $cm^{-1}$ | 1076 | H2O, CO, N20, NO |

Vertical resolution TBD ... (~ 2km)

https://user.eumetsat.int/resources/user-guides/mtg-irs-level-1-data-guide

## InfraRed Sounder IRS (MTG-S)

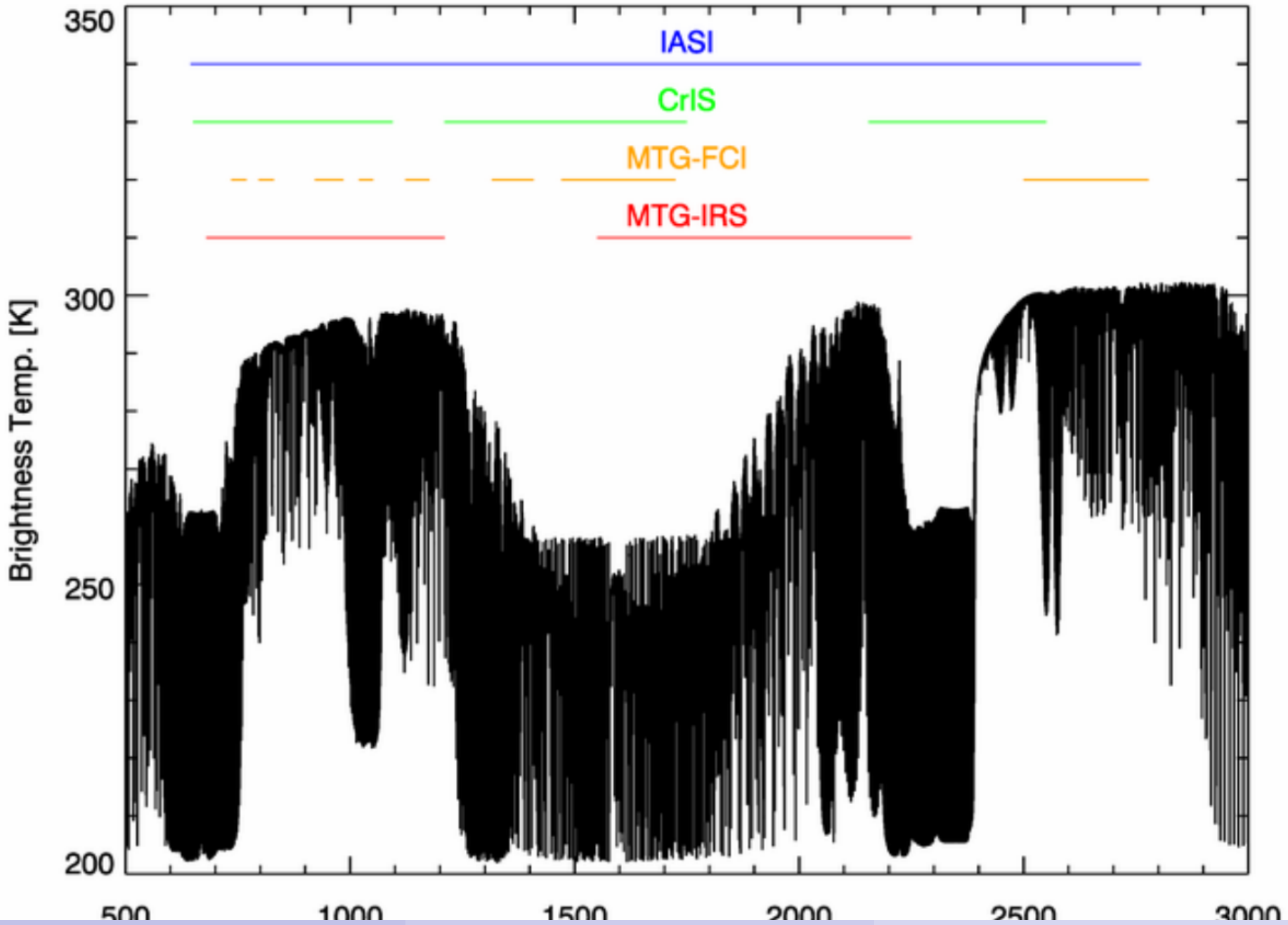
L2 data

- Temperature, humidity and ozone profiles
- Earth Surface parameters
- Instability indices and total vertical column
- Cloud products
- Contextual parameters

Test/Simulated data available here

https://user.eumetsat.int/resources/user-guides/mtg-irs-level-1-data-guide

# Spectral coverage



# Sentinel-4/Ultraviolet, Visible and Near-infrared Spectrometer (S4 / UVN) (MTG-S)

**objective**: Observe the diurnal cycle of the tropospheric composition, trace gases, aerosol and cloud properties
O3 (Ozone), NO2 (Nitrogen dioxide), SO2 (Sulfur dioxide), HCHO (Formaldehyde), CHOCHO (glyoxal)

| | |
|---|---|
| Instrument type | Passive imaging spectrometer |
| Spectrometric bands | **3**, Ultraviolet (305-400 nm), Visible (400-500 nm) and Near Infrared(750-775 nm) [1] |
| Spectrometric channels | **2**, (UV-VIS channel; NIR channel) |
| Field Of View | E-W: 30°W-46.5°E @ 40°N, N-S: 30°N-65°N |
| Spatial resolution | 8 x 8 $km^2$ |
| Spectral resolution | 0.5 nm for the UV-VIS channel; 0.12 nm for the NIR channel |
| Radiometric accuracy (absolute) | 3% (2% goal) of the measured sun irradiance, earth radiance and spectral reflectance. |
| Revisit time | ~ 60 min |

# Sentinel-4/Ultraviolet, Visible and Near-infrared Spectrometer (S4 / UVN) (MTG-S)

L1 data

| Level-1B product | Parameter(s) | Distributed |
|---|---|---|
| Earth Radiance | Spectrally and radiometrically calibrated and geo-located Earth radiance | To all users |
| Solar Irradiance | Spectrally and radiometrically calibrated Solar irradiance | To all users |
| DPPF | Data Processing Parameters | To Expert Users |
| Calibration | Calibration data. | To Expert Users |
| Star | Star calibration data | To Expert Users |
| Diagnostic | Instrument diagnostic data | To Expert Users |

# Sentinel-4/Ultraviolet, Visible and Near-infrared Spectrometer (S4 / UVN)

L2 data

| Level-2 products | Parameter(s) |
|---|---|
| O3 | Ozone (O3) total column, tropospheric sub-column, |
| NO2 | Nitrogen dioxide (NO2) total column, tropospheric sub-column |
| SO2 | Sulfur dioxide (SO2) total column |
| HCHO | Formaldehyde (HCHO) total column |
| CHOCHO | Glyoxal (CHOCHO) total column |
| Cloud | Cloud optical thickness, cloud fraction, cloud altitude |
| Aerosol | Aerosol column optical thickness, type, layer height, absorbing index |
| Surface | Surface and aerosol characteristics |

https://sentinels.copernicus.eu/missions/sentinel-4

## Main data resources

- EUMETSAT data store / EUMETSAT data catalogue
- EUMETSAT 'data archive'
- Satellite Application Facilities SAF
  - AC SAF, Atmospheric Composition Monitoring, ozone, trace gases, aerosols and ultraviolet data
  - H SAF, Operational Hydrology and Water Management, data for operational hydrological
  - LSA SAF, Land Surface Analysis, land, land-atmosphere interactions and biosphere
  - NWC SAF, Nowcasting and Very Short Range Forecasting, forecast for the next few hours
  - NWP SAF, Numerical Weather Prediction, interface between satellite data and European activities in NWP
  - OSI SAF, Ocean and Sea Ice, ocean-atmosphere interface
  - ROM SAF, Radio Occultation Meteorology, high-quality GNSS Radio Occultation (RO) data for NWP and climate application areas

## Main reference resources

- EUMETView, interactive map and product explorer
- EUMETSAT User Portal, guides and products references
  - Training
  - Software
  - Guides
  - .....

## Land Surface Analysis **LSA** SAF available products

- Land Surface Albedo (MTDAL)
- Fire Radiative Power (MTFRP)
- Land Surface Temperature (MTLST)

https://datalsasaf.lsasvcs.ipma.pt/PRODUCTS/MTG/

# Atmospheric Composition Monitoring **AC** SAF available products

**NRT products**

- Total ozone column
- Total and tropospheric NO2 columns
- Total SO2 column
- Total HCHO column
- High-resolution ozone profile
- Absorbing aerosol height
- Absorbing aerosol index from PMDs
- Global 1-day UV index forecast
- Global tropospheric ozone column
- IASI CO
- IASI SO2
- IASI HNO3
- IASI total O3 and O3 profiles

**Offline products**

- Total ozone column
- Total and tropospheric NO2 columns
- Total SO2 column

## Other Links

- MTG in operations
  Current status and operational information
- MTG test data
  If you want to test possible applications

# This was an overview of the MTG satellites and data

*ευχαριστώ! thanks!*

contact: **anatsis@noa.gr**

presentation: **github.com/thanasisn/presentations**